Use Cases for both:

Name	File Name Input
Id	UC_01
Description	Right when the program is started, the user is prompted to enter a filename containing ballots. The user can enter as many files of ballots as they would like, as long as they enter at least 1.
Actors	User / Election Official
Organizational Benefits	Allows the user to implement multiple files containing multiple ballot,s instead of having the user put in each ballot one by one
Frequency of use	Happens at the beginning of the election - repeated for every file the user enters
Triggers	The user starts the program and is immediately prompted to input the name of a file. They are then prompted to move on or to add more files.
Preconditions	The user has started the program, and they have file(s) containing ballot(s)
Postconditions	The system contains every single ballot that will be used to complete the election
Main course	1) The user starts the program 2) The user is prompted to enter the name of the file 3) The system stores the file in 1 location
Alternate course	1) The user starts the program 2) The user is prompted to enter the name of the file 3) The user has more files to be entered 4) The system prompts the user to add files until they are all entered 5) The system stores the files in 1 location
Exceptions	1) The user enters a file name that doesn't match a ballot file 2) The user is alerted to their mistake and given the chance to re-enter the file name

Name	Seat Input
Id	UC_02
Description	After starting the program, the election official will be prompted to enter the number of seats to fill for the election.

Actors	User / Election Official
Organizational Benefits	Allows the election to stay fair and accurate by selecting the correct number of winners in the election.
Frequency of use	Once per election
Triggers	The user is prompted and then inputs a number
Preconditions	The ballots have been entered, so the # of candidates is determined
Postconditions	There is an int number of seats, 0 < n < # of candidates
Main course	1) The user starts the program 2) The user enters 1 or more files containing all ballots 3) The number of candidates is determined based on the ballots 4) The user is asked to provide a number of seats
Alternate course	N/A
Exceptions	1) The user enters either an input that isn't an int or an int that is not, 0 < n < # of candidates 2) They are told their input is invalid and prompted to try again 3) This continues until a correct input is entered

Name	Algorithm Input
Id	UC_03
Description	After starting the program, the election official will be prompted to enter the algorithm to use to decide the winners.
Actors	User/election official
Organizational Benefits	Allows the election to stay fair and accurate by selecting the correct voting algorithm.
Frequency of use	Once per election
Triggers	The user is prompted to input “1” for STV or “2” for Plurality.
Preconditions	The user has already entered 1 or more files containing ballots, so we know there is actually an election to be determined.
Postconditions	A type of election has been determined.
Main course	1) The user starts the program 2) The user enters 1 or more files containing all ballots 3) The user enters the number of seats

	4) The program asks the user to select an algorithm
Alternate course	N/A
Exceptions	1) The user enters any input other than “1” or “2” 2) The program informs the user of the error and prompts them to try again 3) This continues until a correct input is entered

Name	Determine # of ballots and candidates
Id	UC_04
Description	After the election official enters all of the ballots, the program must internally determine the # of ballots and # of candidates.
Actors	System
Organizational Benefits	Allows each candidate to be represented and each vote to count by determining the correct # of both candidates and ballots.
Frequency of use	Once per election
Triggers	The user has declared that all ballots are entered. The ballots and candidates can now be accurately calculated.
Preconditions	The user has entered 1 or more files containing ballots.
Postconditions	There are 0 < n ballots and 1 < n candidates.
Main course	1) The user has entered files of ballots and has declared finished entering files. 2) There is a variable, numCandidates, that can be determined based on ballot 1 by looking at how many options were on the ballot 3) There is a variable numBallots that starts at 0 and increments by 1 for each ballot 4) Go through each ballot in each file to determine the final ballot count
Alternate course	N/A
Exceptions	1) If any files are empty, then disregard the file and move on tothe next without changing either variable

Name	Help Window
Id	UC_06
Description	When the user prompts the program for help, a help window should appear giving information about the program.
Actors	User/Election Official
Organizational Benefits	Allows the user to receive help with the program without having to reach out to a software developer.
Frequency of use	Every time the user needs help with a direction provided by the system.
Triggers	User / Election Official selects help in our User Interface
Preconditions	The user has started the program.
Postconditions	A help window is presented to the user.
Main course	1) The user prompts the program for help 2) The system presents a help window 3) The user can then prompt the program to return 4) The system will go back to the original window
Alternate course	N/A
Exceptions	1) The election is actively running; in this case, the help window will not be propagated to the user.

Name	Election Details Display
Id	UC_05
Description	After the election has been completed, the system will display a summary of the election results on the screen, including election details and the outcome for each candidate.
Actors	System, User / Election Official
Organizational Benefits	Provides an overview of election results, allowing election officials to assess outcomes and statistics
Frequency of use	Once per election run, after the election results are determined
Triggers	The election has finished running, and the winner(s) have been declared
Preconditions	Election results have been computed. Elected and non-elected lists available for the STV Election Election type, number of ballots, seats, and candidates are known.
Postconditions	Election results and statistics are displayed on screen.

Main course	1. Retrieve and display election type (STV or Plurality), number of ballots, number of seats, and number of candidates. 2. Display elected and non-elected candidates with the percentage of votes received, if election type was Plurality.
Alternate course	AC1: Election type was STV 1. Display the order of the winning and losing candidates for STV.
Exceptions	EX1: System fails to retrieve election results 1. System notifies user of the error

Use Cases for Plurality:

Name	Plurality Voting Algorithm
Id	PLU_01
Description	After the user chooses the plurality algorithm in the algorithm input, a plurality algorithm will be used to determine the winner(s) of the election.
Actors	System
Organizational Benefits	Correctly determines the winner(s) of elections by determining the candidate(s) with the greatest number of votes.
Frequency of use	Once per time that the user chooses the plurality algorithm in the algorithm input
Triggers	User chooses “2” (representing Plurality) in the algorithm input
Preconditions	Each ballot will only have one candidate indicated as their choice for the vote. Each ballot that is generated will only have a single 1 in the candidate’s position that was voted for by the voter. The number of seats will have been input by the user. User chooses “2” in the algorithm input.
Postconditions	The winner(s) of the election have been determined using the plurality voting algorithm.
Main course	1. A count is maintained for each candidate, starting at 0 for each 2. Each ballot is checked. 3. If a ballot chooses a candidate, the count for that candidate is increased by one. 4. The candidate(s) with the greatest number of votes will be declared the winner(s).
Alternate course	AC1: There is a tie in counts for a given seat. 1. An electronic “fair coin toss” will determine the winner.

Exceptions	EX1: System fails to execute Plurality algorithm. 1. System notifies user that an error has occurred.

Use Cases for STV:

Name	STV Voting Algorithm (Droop Quota)
Id	STV_01
Description	After the user chooses the STV algorithm in the algorithm input, the Single Transferable Vote (STV) method using the Droop Quota will be applied to determine the elected and non-elected candidates.
Actors	System
Organizational Benefits	Correctly determines the winner(s) of elections by distributing votes according to voter preferences.
Frequency of use	Once per time that the user chooses the STV algorithm in the algorithm input
Triggers	User chooses “1” (representing STV) in the algorithm input.
Preconditions	Each ballot will have at least 60% of the candidates ranked. The number of seats will have been input by the user. User chooses “1” in the algorithm input.
Postconditions	Candidates are placed into Elected and Non-Elected lists.
Main course	1. Shuffle ballots 2. Calculate Droop Quota 3. Distribute ballots into piles according to their first-choice candidate. 4. If any candidate reaches the quota, declare them elected and record their name. Their ballots are removed permanently. Subsequent ballots listing them will transfer to the next preferred candidate. 5. If no new candidate meets the quota, eliminate the candidate with the fewest votes. Redistribute their ballots according to the next preference. 6. Repeat steps 4–5 until all ballots are processed and all seats are assigned. 7. Create Elected and Non-Elected lists, ordered based on when they were elected and when they were placed on the non-elected list, respectively, and provide tothe user
Alternate course	AC1: A Tie occurs when eliminating a candidate. 1. The candidate who was last to receive their first ballot is eliminated 2. Return to Main Course step 5

	AC2: User chooses not to shuffle ballots. 1. Skip Main Course step 1 2. Return to Main Course step 2
Exceptions	EX1: Error occurs when executing the STV algorithm 1. The system notifies the user that an error has occurred.

Name	Generate STV Audit Report
Id	STV_02
Description	At the end of the election, a .txt report will be automatically generated, including information on the type of election, the date, candidate names and parties, number of candidates, seats to be filled, elected list in order, and non-elected list in order.
Actors	System
Organizational Benefits	Allows the user to view a report of the election to check that all information is correct and to have a historical record of the election.
Frequency of use	Once per election at the very end.
Triggers	The election has ended, and the winners have been declared; the report can then be generated.
Preconditions	An election is run and successfully completed.
Postconditions	A .txt Audit report is created and given to the user.
Main course	1) The election is run 2) The winner is declared, ending the election 3) The Report is generated in a .txt file with all the accurate information from the election 4) The report is placed somewhere that the user can access
Alternate course	N/A
Exceptions	N/A

Name	Shuffle Ballot Selection
Id	STV_03
Description	Before the STV election winner(s) can be determined, the election official decides if the ballots should be shuffled.

Actors	System, User/Election Official
Organizational Benefits	Allows election officials to run tests with no shuffle before the real election, with a shuffle, and ensure the system is calibrated correctly.
Frequency of use	Once per STV election
Triggers	STV is the chosen election type
Preconditions	The user chooses “1” (representing STV) in the algorithm input.
Postconditions	The ballots are shuffled randomly
Main course	1) User is prompted whether they want ballots shuffled or not 2) If the user selects yes, the ballots are shuffled
Alternate course	AC1: User chooses not to shuffle 1. Ballots stay unshuffled
Exceptions	N/A